columnar_storage_parquet (Initial Draft)

Domain: Database Systems | Document ID: columnar_storage_parquet_v1
Specification Standard: IEEE Capstone Evaluation Corpus

Executive Overview:

Column-oriented storage formats, dictionary encoding, run-length encoding (RLE), and bit-packing in Apache Parquet.

Methodological Verification & System Lineage:

This specification is catalogued in IntentCloud for intent-aware hybrid retrieval.
Text chunks are indexed into 384-dimensional dense semantic vectors and 1,000,003-dimensional sparse feature-hashed token projections.
Candidate pools are unified using scale-invariant Reciprocal Rank Fusion (k=60).
Explainable citation attribution is extracted using cross-encoder cross-attention.